

Graph Theory & Networks (AI HL)



Basic graph terminology

- 1 Define the following terms as used in graph theory: vertex, edge, degree of a vertex, and simple graph.
 - 2 A graph has vertices A, B, C, D with edges AB, AC, AD, BC, CD . Find the degree of each vertex, and verify the handshaking lemma (that the sum of degrees equals twice the number of edges).
-

Adjacency matrices

- 3 Write the adjacency matrix for the graph with vertices A, B, C and edges AB, BC (undirected, no weights), using the vertex order A, B, C .
-

Walks, paths, and connectedness

- 4 Explain the difference between a walk, a path, and a cycle in a graph.
 - 5 Explain what it means for a graph to be connected, and give an example of a disconnected graph with 4 vertices.
-

Trees and spanning trees

- 6 State the defining property of a tree (as a type of graph), and find the number of edges in a tree with 7 vertices.
-

Applying Kruskal's algorithm for a minimum spanning tree

- 7 A weighted graph has edges with weights: $AB = 4, AC = 2, BC = 3, BD = 5, CD = 6$. Use Kruskal's algorithm (adding the cheapest edge that doesn't create a cycle, in order of increasing weight) to find the minimum spanning tree, and state its total weight.
-

Eulerian and Hamiltonian considerations

- 8 State the condition on vertex degrees for a connected graph to have an Eulerian circuit (a closed walk using every edge exactly once), and explain why the classic "Seven Bridges of Königsberg" graph fails this condition.

The Chinese Postman Problem (route inspection)

- 9 A graph representing a delivery route has exactly two vertices of odd degree, B and D , with the shortest path between them having total weight 7. Explain how to find the minimum-weight closed walk that traverses every edge at least once (the route inspection algorithm), and state the total extra weight that must be added.

Applying Dijkstra's algorithm for shortest paths

- 10 A network has direct edges: $A-B = 4$, $A-C = 2$, $B-D = 5$, $C-B = 1$, $C-D = 8$. Use Dijkstra's algorithm reasoning (always extending from the currently cheapest-reached unvisited vertex) to find the shortest path from A to D , and state its total weight.

Prim's algorithm for a minimum spanning tree

- 11 Using the same weighted graph as before (edges $AB = 4$, $AC = 2$, $BC = 3$, $BD = 5$, $CD = 6$), apply Prim's algorithm starting from vertex A (always adding the cheapest edge connecting a vertex already in the tree to one not yet in the tree) to find the minimum spanning tree, and confirm it matches the result from Kruskal's algorithm.

Isomorphic graphs

- 12 Explain what it means for two graphs to be isomorphic, and state two necessary (but not sufficient) conditions that isomorphic graphs must share.

The travelling salesman problem: nearest-neighbor algorithm

- 13 A network has distances $AB = 6$, $AC = 9$, $AD = 4$, $BC = 3$, $BD = 8$, $CD = 5$. Starting at A , use the nearest-neighbor algorithm (always travel to the closest unvisited vertex) to find an upper bound for the shortest route that visits every vertex once and returns to A , and state its total length.

Planar graphs and Euler's formula

- 14 A connected planar graph has 8 vertices and 12 edges. Use Euler's formula for planar graphs, $V - E + F = 2$, to find the number of faces (regions, including the outer, unbounded region).

Weighted adjacency matrices and shortest paths

- 15** A weighted graph has vertices A, B, C with edge weights $AB = 5$, $AC = 9$, $BC = 2$ (no direct edge is represented by 0 on the diagonal, and a dash for a non-existent edge). Write the weighted adjacency matrix in the order A, B, C , and use it to state the shortest path from A to C .